Practice workbook · Apply the triplet code to bases, codons and amino acids

Converting DNA triplets to mRNA codons and tRNA anticodons, reading amino acids from a table, counting bases per amino acid, and counting possible codes as a power.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-APPLY-TRIPLET-CODE

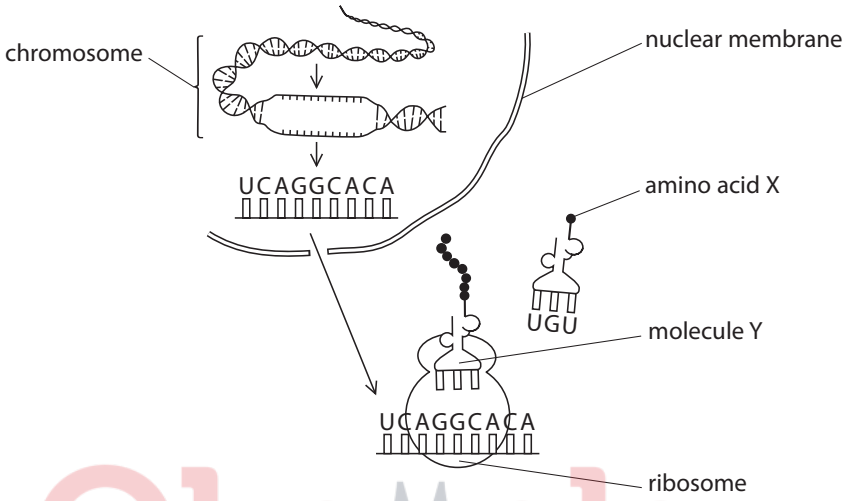
1. Question 6 · part c · 1 mark

June 2019 Paper 1R

Worked-example retry · Attempt part c only.

6 Protein synthesis has a number of stages.

The diagram shows some of the stages.



The table lists the base triplets of DNA that code for some of the amino acids found in a protein.

Base triplet of DNA	Amino acid
AGT	serine
CCG	glycine
TGT	threonine
GTA	histidine
CAA	valine
TCC	argenine
ACA	cysteine

(a) Describe the function of the nuclear membrane.

(2)

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1. Question 6 · part c · 1 mark

June 2019 Paper 1R

Worked-example retry · Attempt part c only.

(b) Which is the correct sequence for the three unpaired bases on the tRNA molecule labelled Y?

(1)

- ☒ **A** AAU
- ☒ **B** CCG
- ☒ **C** GGC
- ☒ **D** UUA

(c) Give the name of amino acid X.

(1)

(d) Describe the process occurring on the ribosome.

(6)

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1. Question 6 · part c · 1 mark · Working and answer

Working and answer

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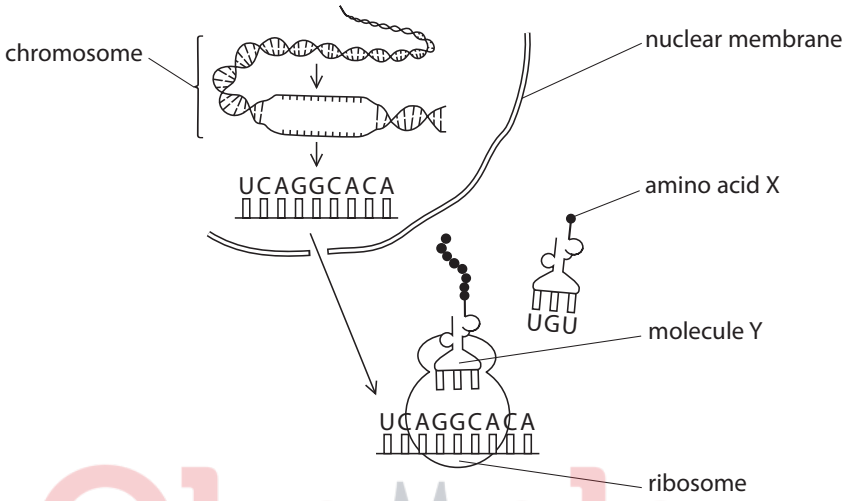
2. Question 6 · part b · 1 mark

June 2019 Paper 1R

Further practice · Attempt part b only.

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2. Question 6 · part b · 1 mark · Working and answer

Working and answer

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3. Question 8 · part a.iv · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.iv only.

8 (a) CFTR is a protein that is found in cell membranes.

This protein is responsible for transporting sodium ions across cell membranes using active transport.

(i) Describe the process of active transport.

(2)

(ii) In some people a DNA mutation causes the production of a faulty CFTR protein.

This results in cystic fibrosis.

Explain how an individual can inherit cystic fibrosis from their parents.

(2)

(iii) The faulty CFTR protein is unable to bind to the endoplasmic reticulum in the cell following protein synthesis.

Suggest how this might affect the CFTR protein.

(2)

(iv) The CFTR protein is made up of 1408 amino acids.

Calculate the number of nucleotides found in one strand of the CFTR gene.

(2)

number of nucleotides = ...

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3. Question 8 · part a.iv · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.iv only.

(v) Describe the structure of a nucleotide found in a DNA molecule.

(3)

(b) There are 20 different amino acids that can be joined to form a polypeptide chain.

The table gives some base sequences that code for five of these amino acids.

Code	Name of amino acid
AAT or AAC	asparagine
TAT or TAC	tyrosine
TTT or TTC	phenylalanine
CGT or CGG	arginine
CAT or CAC	histidine

Diagram 1 shows part of the gene that codes for a CFTR protein that functions normally.

Diagram 2 shows the same part of the gene with a mutation that causes cystic fibrosis.

AATATCATCTTTGGTGTTTCCTATGAT

Diagram 1

AATATCATCGGTGTTTCCTATGAT

Diagram 2

Describe the changes that occur in the gene and the protein that cause cystic fibrosis.

(3)

(Total for Question 8 = 14 marks)

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3. Question 8 · part a.iv · 2 marks · Working and answer

Working and answer

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4. Question 8 · part b · 3 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

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November 2020 Paper 1R

Further practice · Attempt part b only.

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4. Question 8 · part b · 3 marks · Working and answer

Working and answer

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5. Question 1 · part c · 2 marks

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Further practice · Attempt part c only.

1 (a) The table shows some cell structures and their functions.

(i) Place one tick (✓) in each row to show the function of each cell structure.

(3)

Cell structure	Function		
	makes proteins	**contains DNA**	**allows substances in and out of cells**
nucleus			
ribosomes			
cell membrane			

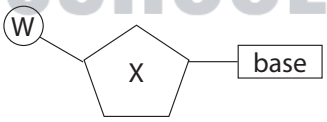
(ii) Give a structure found in bacterial cells.

(1)

.....

(b) (i) DNA is made of two chains of nucleotides.

The diagram shows one nucleotide.



Name part W and part X.

(2)

W .....

X .....

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5. Question 1 · part c · 2 marks

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Further practice · Attempt part c only.

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(ii) Describe how bases hold the two chains of nucleotides together.

(3)

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(c) The gene for the hormone insulin contains 153 bases.

Determine the number of amino acids in insulin.

(2)

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number of amino acids = ...

(Total for Question 1 = 11 marks)

5. Question 1 · part c · 2 marks · Working and answer

Working and answer

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6. Question 5 · part a · 2 marks

November 2023 Paper 1

Further practice · Attempt part a only.

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5 DNA codes for the production of proteins in cells.

(a) The DNA code is read, three bases at a time. There are four different bases.

Calculate how many different three base codes are available.

(2)

number of codes = ...

(b) (i) Explain how a mutation in DNA can cause a change in the phenotype of a cell.

(3)

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(ii) State **one** factor that can cause mutation.

(1)

(iii) A mutation can change the shape of an enzyme molecule.

Explain why this may change how the enzyme works.

(2)

6. Question 5 · part a · 2 marks · Working and answer

Working and answer

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